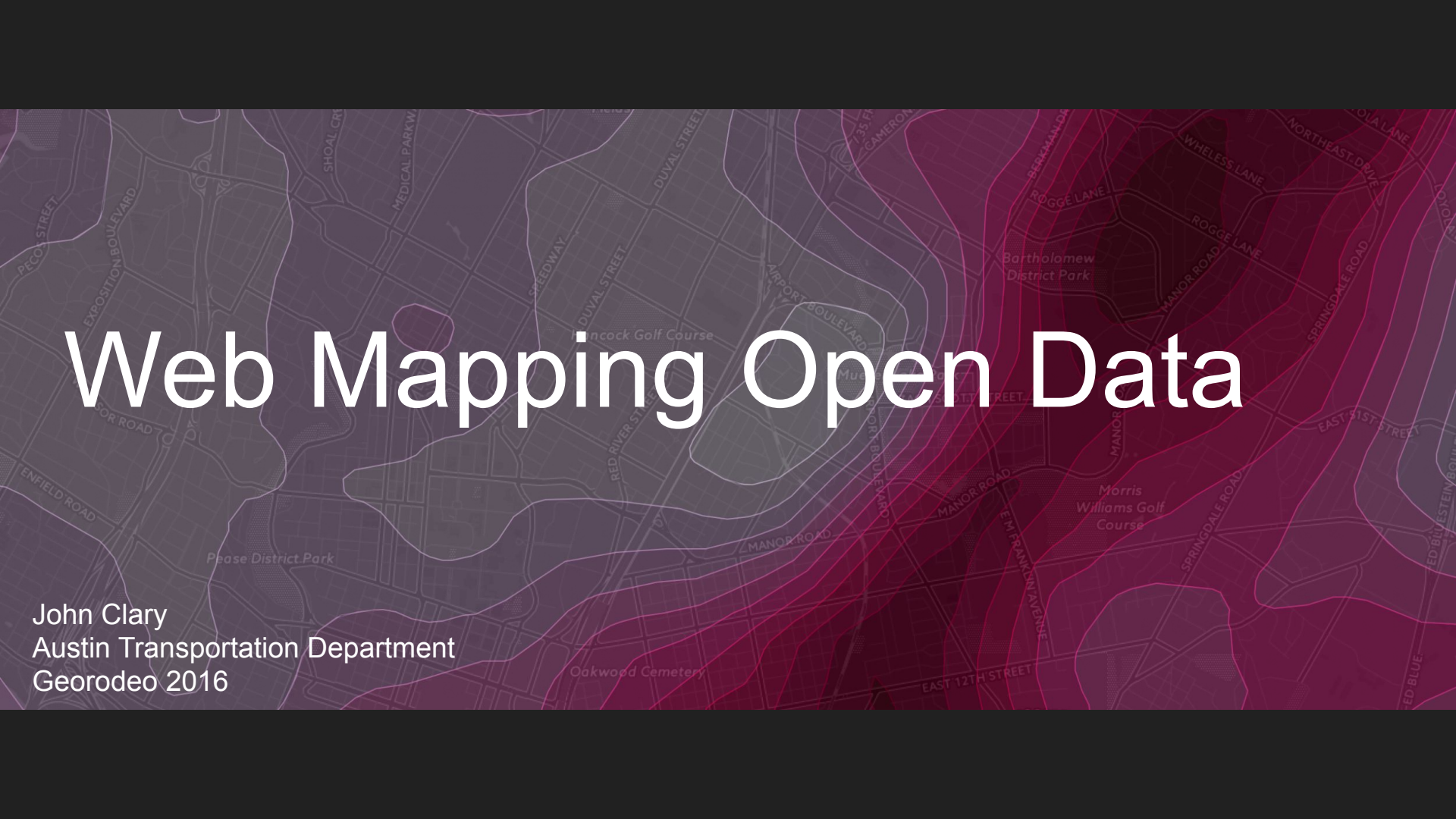


A topographic map of a region in Austin, Texas, featuring a color gradient from dark purple to red. The map shows a network of streets, including Exposition Boulevard, Duval Street, Airport Boulevard, Manor Road, and Springdale Road. Key landmarks such as Incocks Golf Course, Pease District Park, Bartholomew District Park, and Morris Williams Golf Course are labeled. The title 'Web Mapping Open Data' is centered in large white text.

Web Mapping Open Data

John Clary
Austin Transportation Department
Geordeo 2016

If you want to participate

Slides and data: spatialaustin.com/georodeo

Create a [GitHub](#) account

Get yourself [a text editor](#)

Saddle up!

A Gander at the Web Mapping Landscape

Web Mapping Tips for City Slickers

Yeehaw! Map Wranglin'!

spatialaustin.com/georodeo

GIS and Open Data

GIS developers have played an important role in the free and open source software movement

GIS users are key creators and consumers of open data

Open data and the geospatial web are evolving in tandem

Managing and mapping open spatial data should be a core competency of GIS professionals

Out-of-the-box

Google My Maps

ArcGIS Online

CartoDB

Tableau

Assembly Required

MapBox

LeafletJS

OpenLayers

D3.js

Mapzen

Google, Bing, etc.

Open (Spatial) Data Challenges

Source

Format

Size

spatialaustin.com/georodeo

Open (Spatial) Data Sources

[OpenStreetMap](#)

[Natural Earth](#)

Jurisdictional Open Data Portals (e.g. [Austin](#))

[US Open Data Census](#)

[ArcGIS Online](#)

Open (Spatial) Data Formats

GeoJSON

Widely Accepted

Simple spec*

Tolerable File Size

TopoJSON

Widely Recognized

Complex spec

Small File Size

Ask Calvin Metcalf
(he's here!)

Preparing data for the web

**TAKE ONLY WHAT YOU NEED TO
SURVIVE**



Leave behind

Extraneous fields

Precision

Accuracy

Overlapping geometry

City of Austin Council Districts

GeoJSON: 1,400kb

Dropping extraneous fields

Desktop GIS

Append a field name filter to your Socrata URL:
[?\\$select=single_mem,the_geom](#)

Mapshaper

Try: `mapshaper -filter-fields council_di -o`

So long, precision!

```
[-97.5283229444848, 30.27364793546887]
```

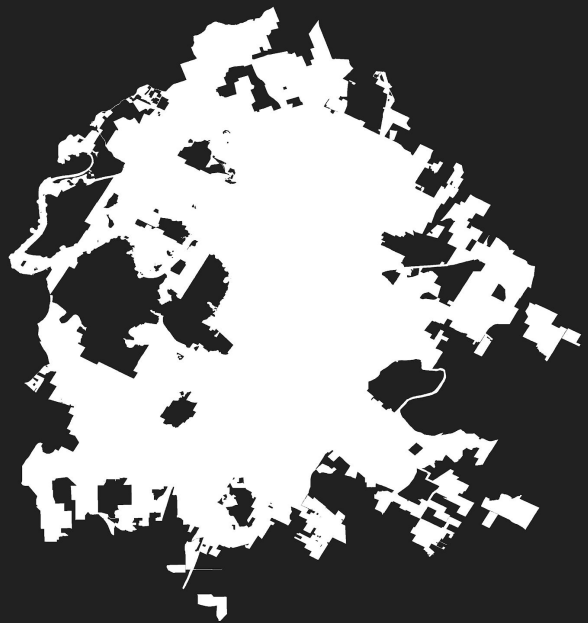
In mapshaper try:

```
mapshaper -o -precision=.00001
```

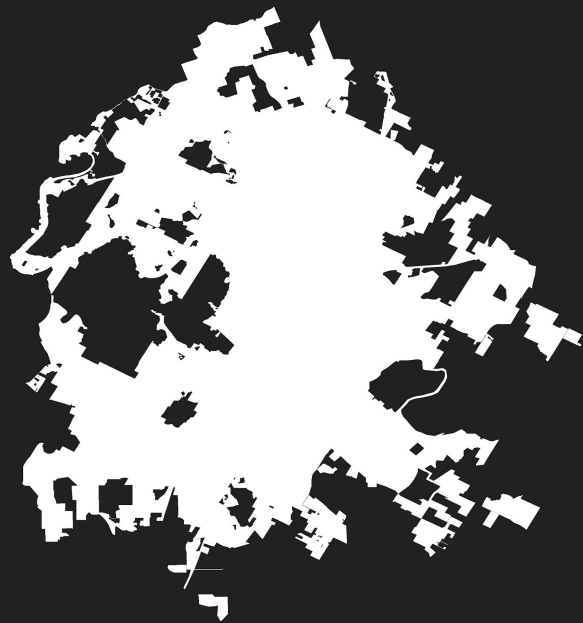
So long, accuracy!

Use mapshaper to simplify to 12%

1,200kb



78kb





1,200kb

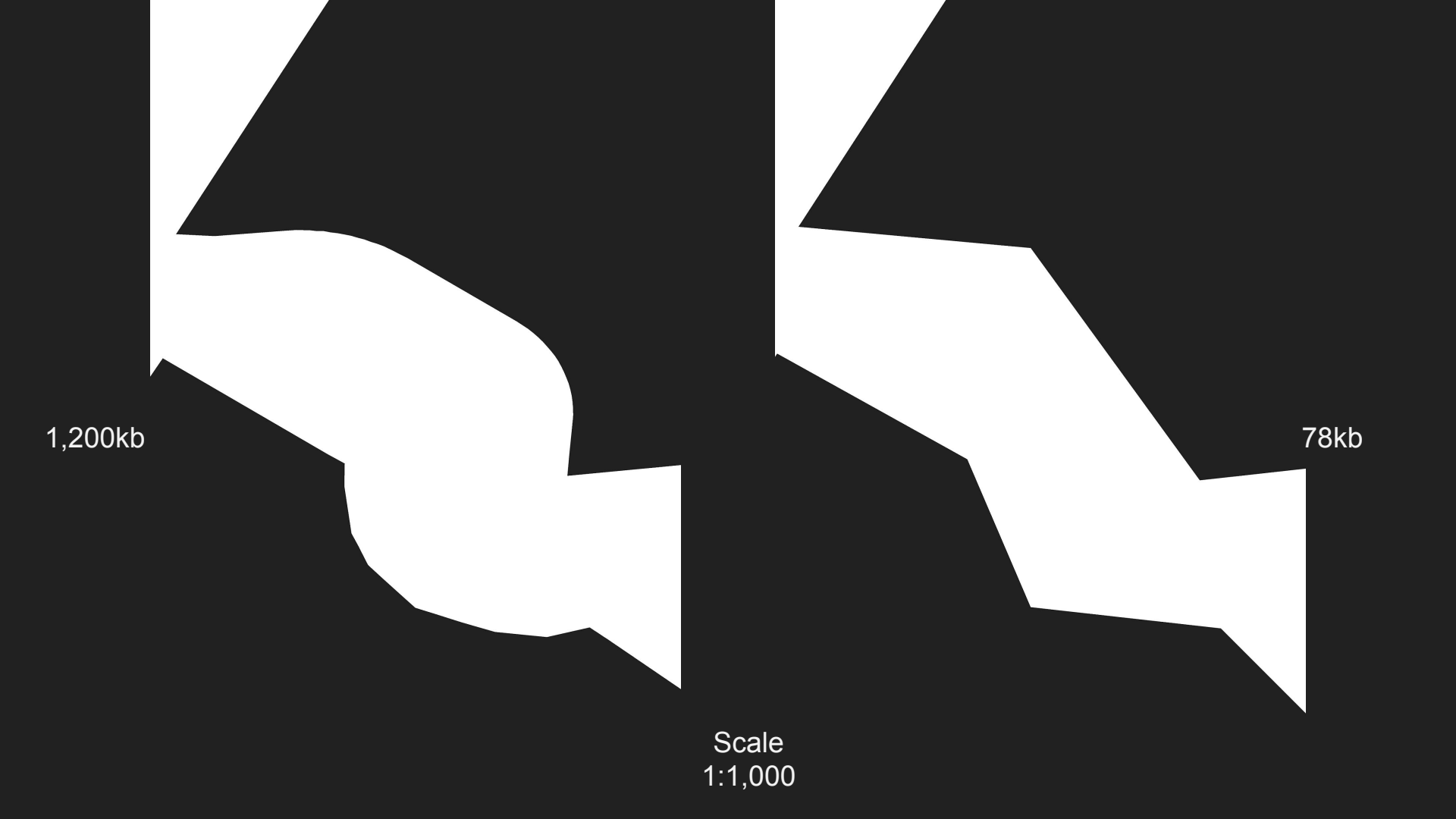
This is an aerial photograph of a coastal region, likely a bay or inlet. A red line is drawn across the map, starting from the top right, curving around a headland, and then following the shoreline. The map shows various land features, including what appears to be a large industrial or commercial area with several buildings and a parking lot. The water is a dark blue-grey color. The red line is approximately 1,200kb in length.



78kb

This is a zoomed-in aerial photograph of the same coastal region shown in the left panel. The red line is more clearly visible, showing its path along the shoreline and around the headland. The map shows more detail of the land features, including the buildings and parking lot. The water is a dark blue-grey color. The red line is approximately 78kb in length.

Scale
1:50,000



1,200kb

78kb

Scale
1:1,000

1,200kb



78kb



Other Spatial Data Mgmt Tools and Techniques

Vector tiles! (Raster tiles, too)

Geojson.io

GDAL / OGR via Sara Safavi

QGIS

Yeehaw!

Yeehaw! - Wranglin' Maps with LeafletJS

Where many Austinites have gone before:

[James Seppi](#)

[Sara Safavi](#)

[Preston Pham](#)

Step 1

Download and unzip data:

spatialaustin.com/georodeo

Open index.html with your editor, and also with your browser

Step 2

Remove leading slashes to add layers to the map:

`// .addTo(map);` becomes `.addTo(map);`

Save the file and reload it in your browser

Esri Leaflet (Travis County)

Line 57

Ask John Gravois (He's literally here right now!)

Leaflet GeoJSON (ATX Council Districts)

Line 64

CartoDB.js (Census Tracts)

Line 91

Ask Stuart Lynn (He's here right now!)

Layer Control Selector

Line 118

Austin 311: Lost Item in Storm Drainage System

This might break your browser

Line 79

Turf.js - Joining Service Requests to Polygons

Line 95

Hosting A Map w/ GitHub Pages

1. Join GitHub and verify your email address
2. Create a new repository
3. Drag and drop the contents of your local folder to the repository
4. Create a new branch called 'gh-pages'
5. Go to:
`[your-github-username].github.io/[your-repository-name]`

Remember to attribute your data and code!

Get help and help others

[MaptimeATX](#)

[ATX OSG](#)

[Open Austin](#)

[Austin D3.js Meetup](#)

Thanks and Stay in Touch!

John.Clary@austintexas.gov

@SpatialAustin